

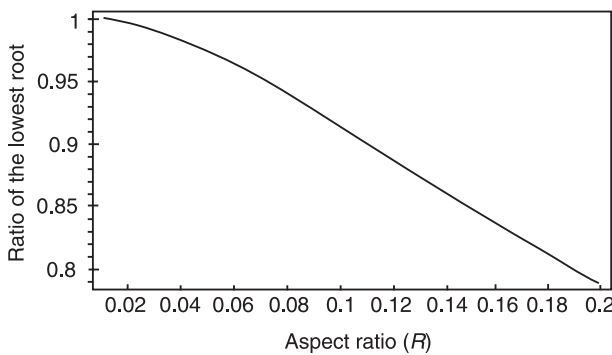


Figure 3 Timoshenko/Euler lowest roots vs aspect ratio.

Nomenclature

a	radius
A	cross-sectional area
E	Young's modulus
G	shear modulus
I	inertia
M	bending moment
R	aspect ratio
v	transfer deflection
V	shear force

ψ	bending slope
ρ	density
κ	Timoshenko shear coefficient

See Plate 8.

See also: Finite element methods; Shock.

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BEARING DIAGNOSTICS

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Roller Bearing

Figure 1A illustrates the cross-section of an angular contact ball bearing. It shows the angle of contact, and the diameters of outer race, rollers, inner race and pitch circle. **Figure 1B** illustrates the cross-section of a taper roller bearing. It shows the angle of contact, and the diameters of outer race (cup), inner race (cone), and rollers. The primary excitations of vibration are the contact forces between the rollers and the raceways, and the rollers and the cage. The frequencies of these contacts are directly related to the roller passing frequency on the raceways, the roller spinning frequency, and the cage spinning frequency which can be theoretically estimated with the equations in **Table 1**.

Failure Modes

Bearing failure modes include corrosion, wear, plastic deformation, fatigue, lubrication failures, electrical damage, fracture, and incorrect design. Out of these, classical localized defects result from fatigue, and certain distributed defects have fairly distinct vibration patterns which will be detailed below. However, the vibration patterns of other failure modes are, in general, not easy to predict, let alone detect and diagnose at onset. While some of them do eventually lead to localized defects, others give almost no vibration warning and therefore call for means other than vibration analysis.

Localized Defects

The most common failure mode of a properly installed and operated rolling element bearing is localized defects such as fatigue spall, which occur when a sizable piece of material on the contact surface is dislodged during operation, mostly by fatigue

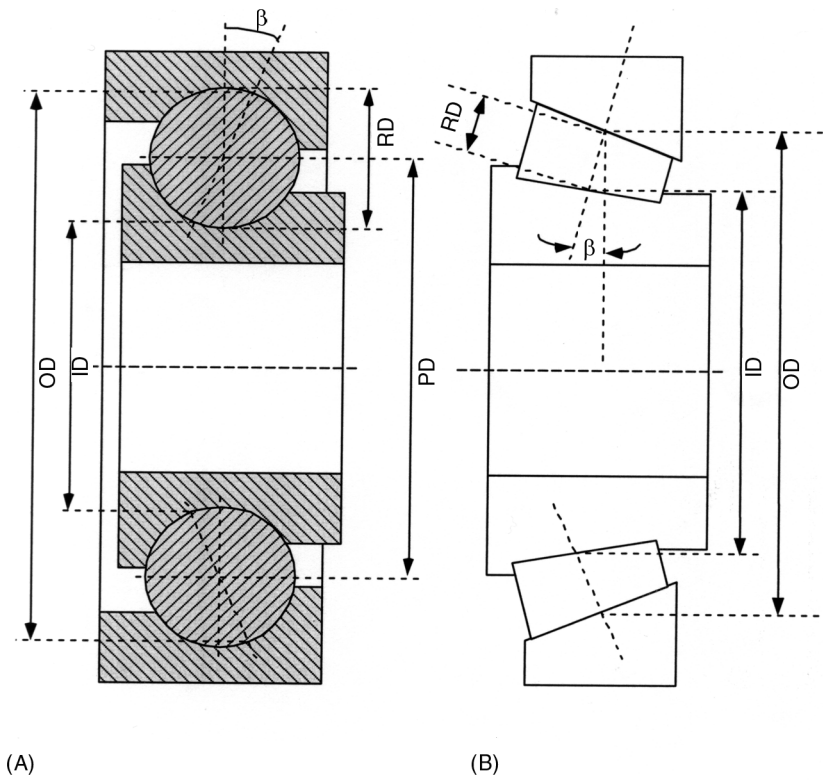


Figure 1 Cross-section. (A) Ball bearing; (B) taper roller bearing.

cracking under cyclic contact stressing. In addition, the bearing cage can fail due to overload or poor lubrication. Thus, a failure alarm for a rolling element bearing is often based on the detection of the onset of localized defects.

Signature-generating Mechanism

During bearing operation, wide-band impulses are generated when rollers pass over a defect at a frequency determined by shaft speed, bearing geometry, and defect location. The frequency, which is often termed characteristic defect frequency, can be esti-

mated as f_{rpi} , f_{rpo} and $2xf_{rs}$ (Table 1) for an inner race, outer race and roller defect, respectively. Some of the vibration modes of the bearing and its surrounding structure, or even the sensor, will be excited by the periodic impulses, and a distinct signature will be generated by such ringings. The leading edge of each ringing usually comprises a sharp rise that corresponds to the impact between a roller and the defect. The ringing then decays due to damping. (It is like driving over a pothole.) Therefore, the signature of a damaged bearing consists of exponentially decaying vibration ringings that occur quasiperiodically at the

Table 1 Important bearing frequencies

	Ball bearing	Taper roller bearing
Inner race roller passing frequency (f_{rpi})	$\frac{n(f_o - f_i)(1 + (RD/PD) \cos \alpha)}{2}$	$n(f_o - f_i) \frac{OD}{OD + ID}$
Outer race roller passing frequency (f_{rpo})	$\frac{n(f_o - f_i)(1 - (RD/PD) \cos \alpha)}{2}$	$n(f_o - f_i) \frac{ID}{OD + ID}$
Cage rotating frequency (f_c)	$\frac{f_i(1 - (RD/PD) \cos \alpha)}{2} + \frac{f_o(1 + (RD/PD) \cos \alpha)}{2}$	$\frac{f_i \times ID + f_o \times OD}{OD + ID}$
Roller spinning frequency (f_{rs})	$\frac{(f_o - f_i) PD}{2 RD} (1 - ((RD/PD) \cos \alpha)^2)$	$(f_o - f_i) \frac{ID}{RD} \frac{OD}{OD + ID}$

n = number of rollers, f_o = outer race rotating speed (rps), f_i = inner race rotating speed (rps), OD = outer race diameter, ID = inner race diameter, RD = roller diameter, PD = pitch circle diameter, α = contact angle.

These equations were derived under the assumption that it is pure rolling contact (no slipping, skidding, etc.) and perfect geometry even under load. One would expect some deviations under actual conditions.

characteristic defect frequency. **Figure 2** shows an example of bearing vibration which has the distinct pattern of periodic ringings. It is sometimes considered as the resonance being amplitude-modulated at the characteristic defect frequency. However, this pattern will begin to fade and become more random as localized defects develop around the races and the rollers. (Then, it is like driving a car on a gravel road.)

The above model does not include many factors such as: radial loading, variation due to transmission path between the signal source and the transducer, misalignment, unbalance, pre-load, and manufacturing defects.

The dominant effect of radial loading is modulation. Assuming a stationary outer race, an inner-race defect rotates around at shaft frequency. Under a radial load, the bearing vibration will be modulated periodically at shaft frequency as the defect goes in and out of the loading zone. For a defect on a roller revolving at the cage frequency, a radial load will result in a modulation at cage frequency. For a defect on the stationary outer race, no modulation will be produced.

Cage fault is different from other localized defects in its signal generating mechanism. It does not usually produce sharp impacts that excite narrow-band bearing resonance. When a cage develops a weak point such as deformation or breakage, a modulation of the wide-band bearing vibration at the cage's rotating speed is common.

Diagnostic Algorithms

Statistical parameters When periodic ringings due to a localized defect become dominant, bearing vibration becomes impulsive. The following statistical parameters have been used as diagnostic parameters for detecting bearing localized defects. For a discrete vibration signal, $x(n)$, let N and $\bar{x}$ denote the length and the mean, respectively. Then:

$$\text{Peak-to-valley} = \max(x(n)) - \min(x(n)) \quad [1]$$

$$\text{RMS} = \sqrt{\left(\frac{1}{N} \sum_{i=1}^N (x(i) - \bar{x})^2\right)} \quad [2]$$

$$\text{Crest Factor} = \frac{\text{Peak to valley}}{\text{RMS}} \quad [3]$$

$$\text{Kurtosis} = \frac{\frac{1}{N} \sum_{i=1}^N (x(i) - \bar{x})^4}{\text{RMS}^4} \quad [4]$$

probability density function

$$p(x \leq x(n) \leq x + \Delta x) = \frac{\text{No. of } x(n) \text{ between } x \text{ and } (x + \Delta x)}{N} \quad [5]$$

A common drawback of these time domain techniques is that they cannot provide information about the location of a defect because they do not take advantage of the characteristic defect frequencies. While the peak-to-valley value is expected to increase due to the periodic ringings, a single noise spike can throw it off and it is also sensitive to changes in operating conditions such as loading and speed. The root mean square (RMS) is not sensitive at the initial stage of damage because a few short-lived, not very large ringings would not change the overall RMS much and operating conditions also affect it. The crest factor provides a normalized peak-to-valley which should be somewhat less sensitive to changes in operation conditions. Kurtosis' quadruple power emphasizes larger amplitudes in defect-induced ringings and therefore is more sensitive to bearing defects than the RMS. The probability density function is expected to have larger tails for a damaged bearing. However, kurtosis, crest factor, and probability density function will fall back to undamaged levels as localized defects spread around races and rollers.

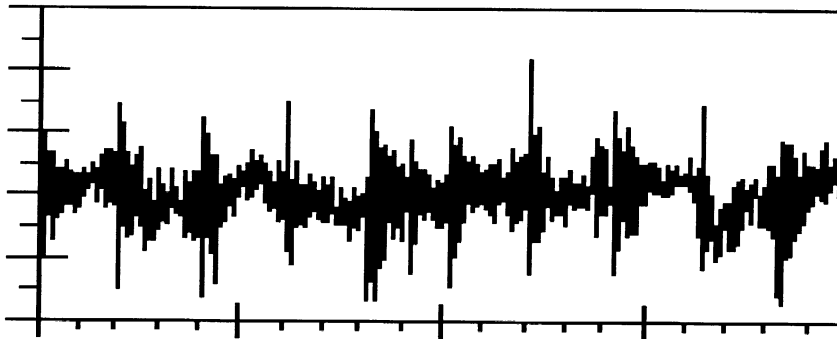


Figure 2 Vibration signal of a bearing with an outer-race defect.

In general, the success of these parameters depends on finding a frequency band that is dominated by bearing ringings. Otherwise, they indicate a change in something other than the condition of the bearing. It was reported that good results can be obtained from higher frequency bands such as 10–20 kHz and 20–40 kHz, where the structure resonance is usual. While it is easy to find some band that works to some extent, the optimal result is difficult to obtain because there is no simple way to predict a bearing structure's natural frequencies, and which one will be excited.

Although trending can be used with these statistical parameters, some of them have been touted as single-shot diagnostic variables that have an absolute threshold and therefore do not need historical data. For example, kurtosis is said to have a value of about 3.0 for a relatively random vibration from an undamaged bearing. Any value significantly higher than 3, say 4.5, is therefore considered as a sign of a bearing fault. Crest factor and probability density function are sometimes considered as single-shot variables. Realistically, these variables are just too simple to offer single-shot diagnosis under all circumstances.

Shock pulse counting This method takes advantage of transducer resonance. When a roller runs over a localized defect, the impact produces a pressure wave. When this pressure wave reaches the transducer, it frequently rings the resonance of the transducer. The number of vibration peaks above a certain threshold during a fixed length of time is then used as a damage indicator. Typically, the transducer is selected to have a resonance frequency between 25 and 40 kHz to avoid the usual machine noise. Due to its simplicity, it is implemented in a couple of off-the-shelf bearing monitoring systems. However, there are few theoretical guidelines about setting the threshold and it has

limited use when there are sources of shock pulses other than a roller bearing. Furthermore, as a time domain method, it cannot determine the location of the defect.

High-frequency resonance technique (HFRT) This is probably one of the best known bearing diagnostic algorithms: it is also known as amplitude demodulation, demodulated resonance analysis, and envelope analysis. The technique band-pass filters a bearing signal to remove low-frequency mechanical noise and then estimates the envelope of the filtered signal. (Enveloping can be accomplished with a rectifier followed by a smoothing operation, or a squaring operation followed by an analog or digital low-pass filter. In addition, the Hilbert transform can also be used if a computer is available.)

The periodicity of the envelope signal, like the one shown in **Figure 3**, is then estimated by spectral analysis or autocorrelation, and compared with the characteristic defect frequencies. If a match is found, the bearing is declared to be damaged.

Obviously, this technique will give the best result if the passing band is chosen to include one or more excited resonances. For simple machines, good results can often be obtained with a fixed passing band with low cut-off placed between 5 and 10 kHz and high cut-off between 25 and 40 kHz. For complex machines such as helicopter transmission, selecting a passing band to include resonances and avoid gear meshing harmonics may not be easy. Varying speed, and therefore meshing harmonics, further compounds the problem.

Synchronized averaging This algorithm can be used as a preprocessing technique to enhance the signal-to-noise ratio. The technique consists of first

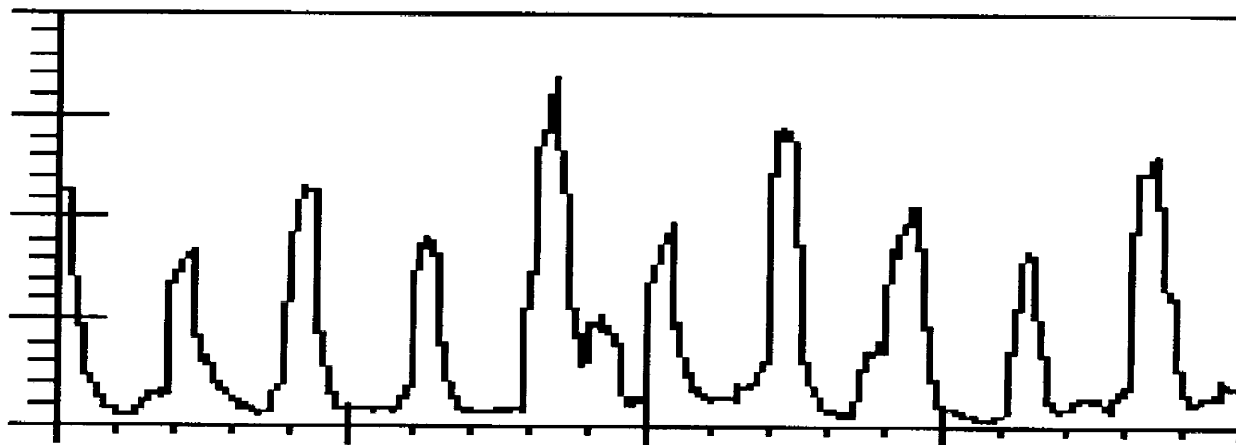


Figure 3 Envelope of the band-pass-filtered signal in **Figure 2**.

calculating the period of repeated ringings by inverting the characteristic defect frequency and then ensemble averaging consecutive segments of a bearing vibration, each one period long. Signals that cannot fit full waves in the period will be attenuated while signals that do will remain. In the original work, only the RMS of the average was calculated. Other diagnostic algorithms can certainly be applied as appropriate.

Cepstral analysis As shown in Figure 2, localized defects produce periodic ringings. The spectrum of such a vibration contains a characteristic defect frequency and its harmonics with the larger magnitude around the excited resonance(s). Since the energy of the bearing vibration is spread across a wide frequency band, it can easily be buried by noise. Therefore, spectral analysis will not be effective. The cepstrum, which was initially defined as the ‘power spectrum of the logarithmic power spectrum’, is useful in detecting spectrum periodicities, such as families of harmonics of bearing defect frequencies, by reducing a whole family of harmonics into a single cepstral line:

$$x_c(\tau) = F\left[\ln|X(\omega)|^2\right] \quad (\text{original definition})$$

or:

$$= F^{-1}[\ln|X(\omega)|] \quad \text{where } X(\omega) = F(x(t)) \quad [6]$$

Cepstral analysis has been shown to be very effective in bearing diagnosis. However, it has not received much acceptance, perhaps because it is more difficult to compute and interpret than the output of, say, the HFRT.

Wavelet transform For a continuous signal $x(t)$, the wavelet transform (WT) is defined as:

$$W_x(a, b) = \int g^{*(a,b)}(t)x(t) dt \quad [7]$$

where $*$ denotes the complex conjugate, and $g(t)$ represents the mother wavelet, e.g.:

$$g(t) = \exp(-\sigma t) \sin(\omega_0 t) \quad \text{for } t \geq 0$$

and:

$$g(t) = -g(-t) \quad \text{for } t < 0 \quad [8]$$

$$g^{(a,b)}(t) = \frac{1}{\sqrt{a}} g\left(\frac{t-b}{a}\right)$$

where a is the dilation parameter which defines a baby wavelet for a given value, and b is the shifting parameter.

For a given a , carrying out WT over a range of b is like passing the signal through a filter whose impulse response is defined by the baby wavelet. Therefore, one may consider WT as a bank of band-pass filters defined by a number of a 's. The salient characteristic of the WT is that the width of the passing band of the filters is frequency-dependent. Therefore, the WT can provide a high-frequency resolution at the low-frequency end while maintaining good time localization at the high-frequency end. This is advantageous for processing transient bearing ringings.

When applied to a bearing signal as a preprocessing tool, the passing band of one or more of the filters could overlap with some of the resonances that are being excited by the roller defect impacts. This results in an enhanced signal-to-noise ratio. For example, Figure 4 shows a bearing vibration measured from a roller-damaged bearing. (Note that it is already high-pass-filtered.) The periodic ringings are not obvious. Figure 5 is the result of WT with one of the baby wavelets. The periodic ringings can be seen more readily and therefore easily identified by, say, an envelope analysis.

By breaking up a broad-band bearing signal into a number of narrow-band subsignals and then scanning them for evidence of bearing defect, WT avoids the risk of selecting a wrong band that does not include any resonance and then missing the defect. The price is that one has to repeat the same bearing diagnostic algorithm on more than one subsignal.

Time-frequency distribution for bearing monitoring The periodic ringings can be revealed or accentuated with time-frequency distributions which give an account of how energy distribution over frequencies changes from one instant to the next. Examples of such distributions include the spectrogram (short-time Fourier transform), Wigner-Ville distribution, and Choi-Williams distribution (CWD).

Figure 6 is a CWD of a bearing signal. Horizontal stripes correspond to the periodic ringing due to a ball passing on an outer-race defect. It is clear that the resonance occurs at 10, 18, and 22 kHz. When compared to WT, which has a better time localization of the ringing, CWD gives a better frequency resolution at high frequencies at a higher computational cost.

The drawback of time-frequency distributions is the medium-to-high computational cost and two-dimensional image-like output, which is usually troublesome for a computer to interpret.

Bicoherence As shown in Figure 2, localized defects produce repetitive ringings. The spectrum of such a vibration contains a characteristic defect frequency and its harmonics, with the largest magnitude around



Figure 4 Bearing vibration with inner-race defect.



Figure 5 Wavelet transform of the vibration in **Figure 4**.

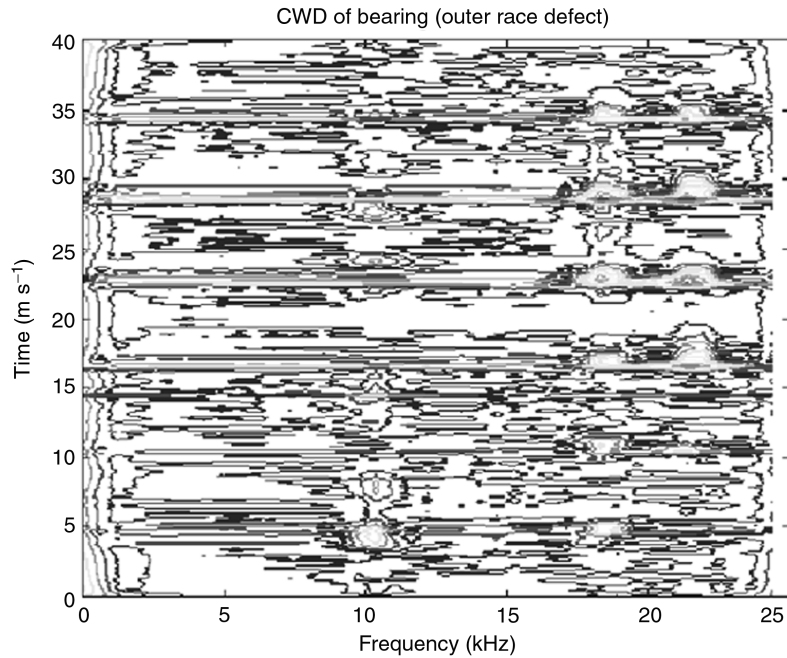


Figure 6 Choi-Williams distribution of bearing vibration with outer-race defect.

the excited natural frequencies. These harmonics are the Fourier components of a periodic signal, and therefore have a fixed phase relationship with one another.

Bicoherence, defined below, quantifies the phase dependency among the harmonics of the characteristic defect frequency, which increases when a bearing is damaged:

$$b^2(f_1, f_2) = \frac{|E[X(f_1)X(f_2)X^*(f_1 + f_2)]|^2}{E[|X(f_1)X(f_2)|^2]E[|X(f_1 + f_2)|^2]} \quad [9]$$

where $E[\cdot]$ denotes the expected value, and $X(f_i)$ denotes the complex frequency component from the Fourier transform. In this case, f_i would be one of the harmonics of the characteristic defect frequency. For example, f_1 and f_2 can be chosen as f_{rpi} and $2f_{rpi}$.

Because bicoherence is based on phase, it works even when the ringings are less than dominant. For example, **Figure 7** shows bearing vibration signals from a good bearing and another with a damaged roller. Although one cannot see the repeated ringing patterns in the bad bearing signal, bicoherence plots can tell them apart (**Figure 8**). While bicoherence outperforms traditional algorithms such as HFRT, it

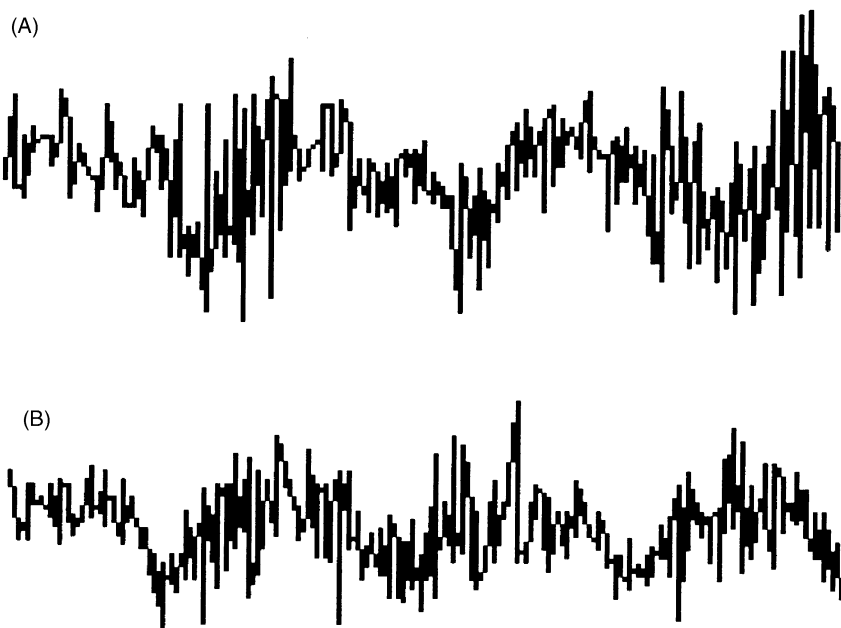


Figure 7 Bearing vibrations. (A) Good bearing signal; (B) bad bearing signal.

is one of the most expensive algorithms in terms of computation cost and data length.

Cage fault When a cage develops a weak point such as deformation or breakage, a modulation of the wide-band bearing vibration at the cage's rotating frequency is common. Traditional bearing diagnostic algorithms such as narrow-band envelope analysis are not effective in detecting this kind of wide-band modulation. One way to determine if a modulation is happening at the cage rotating frequency is to perform a synchronized average, at the cage rotating period, on the envelope of the bearing vibration. The average is then examined for once-per-cage-rotation phenomena.

Alternatively, bandwidth-weighted demodulation can be used.

Summary of localized defect diagnostic algorithms

A significant number of the afore-mentioned algorithms look for the distinct pattern of repeated ringings (Figure 2) associated with the onset and the early stage of damage. Consequently, they suffer from the limitation that they will fall back to the undamaged levels as localized defects spread and the distinct pattern fades away as rollers strike different local defects almost simultaneously all the time. In other words, they do not trend well while damage evolves and this is a problem for the prognosis of bearing life. In addition, since a resonance excited at the onset of a defect may no longer be excited at a later stage, using a band-pass filter to zoom on to a resonance can have its downside of missing another defect-excited resonance later.

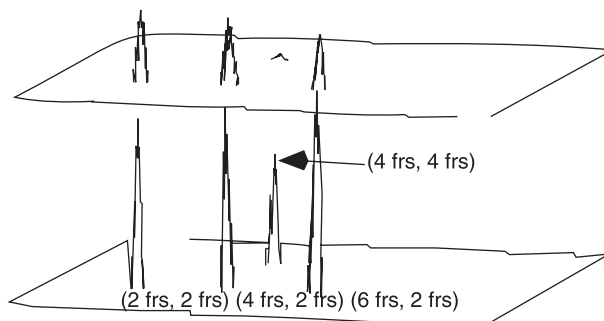


Figure 8 Bicoherence values of good bearing and roller-damaged bearings at four frequency pairs.

Distributed Defects

In addition to localized defects, distributed defects such as off-size rollers and waviness of components are another class of bearing problems. While distributed defects are not generally failures *per se*, they often lead to excessive contact forces and vibrations which in turn lead to premature failures. Fortunately, analysis of vibration spectra is often adequate to diagnose this kind of faults.

A bearing without any distributed defect is considered first. Under a thrust load, roller forces act on the races to produce a deflection. **Figure 9** shows how roller forces act on the inner race. As the rollers and races rotate, the forces and the deflection become periodic. Thus, a vibration sensor located at the inner race would sense vibrations at f_{rpi} and its harmonics. Similarly, a sensor at the outer race would sense vibration at f_{rpo} and its harmonics. **Table 2** gives the frequency associated with a number of conditions while the inner race is stationary.

Table 2 Frequencies associated with various distributed defects

Conditions	Frequencies (sensor at inner race)
One roller off size ($k: 1, 2, \dots$ etc.)	$k(f_{rpi}/n)$
Misaligned outer race	Side band $k(f_{rpi}/n) \pm f_o$
Waviness in outer race ($m =$ number of full waves around the circumference)	Side band $k(f_{rpi}/n) \pm mf_o$

From Meyer *et al.* (1980).

Journal Bearing

Figure 10 shows the three basic parts to a journal bearing: the outer housing, the journal, and the lubricant. The outer housing is a cylindrical shaft, which contains a hollow core large enough to create a close fit between itself and the journal. In addition to confining the path of the journal's orbit, it provides radial support to the journal through direct contact or by aiding in the creation of the oil wedge. A small clearance space between the outer housing and the journal is necessary: in order to assist in the assembly of the journal and the bearing, to provide space for the addition of the lubricant, to accommodate thermal expansion of the journal, and to anticipate any journal misalignment. Within this small clearance space resides the lubricant which provides the basic function of lubricating the journal and outer housing contact, as well as producing the load-carrying capability of the journal bearing and possibly attenuating the vibrations of the rotors. This load-carrying capability is a result of the pressure that is developed by

the viscous effects within the thin film lubricant. A more complex journal bearing that is commonly used is the tilt-pad journal bearing. This type of journal bearing contains the same components as the simple bearing, with the added feature of tilt pads. Tilt pad journal bearings are used for their innate ability better to handle rotor-dynamic instability problems; however, they provide less damping than the simple bearing. Consequently, if vibrations occur due to other sources aside from the bearing, then the amount of damping provided by the bearing may arise as an issue.

Diagnostic Algorithms

Unlike roller bearings, whose vibration has simple and distinct patterns that can mostly be predicted from the geometry, journal bearings, while simple-looking, are rather complicated in their dynamics. Since journal bearings are only one part of a rotating machine, the analysis of the bearings must take into consideration vibrations caused by the flexible journal, fluid and rotor dynamics, and any outside sources of vibration acting on the system. Consequently, this type of analysis usually becomes complicated, and results in no clear vibration patterns associated with most of the failure modes. This is why the tools for the diagnosis of a journal bearing stagger are lacking.

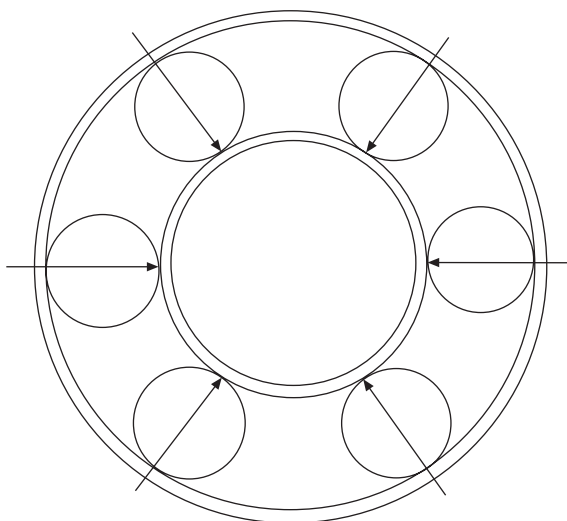


Figure 9 Roller forces acting on inner race.

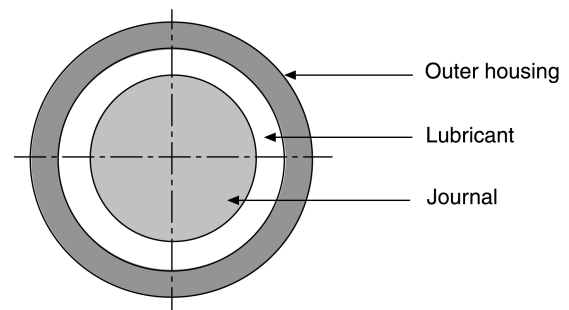


Figure 10 Simple journal bearing.

‘Whirl’ refers to a circulating path the journal takes within the bearing resulting from large vibrations of the journal. Proximity sensors can be placed at perpendicular positions to the orbit of the shaft. These sensors would allow the monitoring of the position of the orbit, and notify the user of the misalignment, the amount of clearance existing, and the range of motion of the journal. The user would be warned if the motion of the journal exceeds a certain allowable envelope of motion.

An accelerometer can be placed on the outer housing to measure the frequency and amplitude of the housing vibrations which do not have a straightforward relationship with journal whirl. Based on the vibrations taken from the bearing’s normal conditions, abnormal conditions can be detected. For example, in cases where the bearing’s housing is supported on springs rather than secured directly to an immobile object, chaotic motion is found at intermediate speed ranges, and disappears at low and high speeds. At low and high speeds, different distinct subharmonic frequency components in the X- and Y-direction of the bearing are excited. However, during intermediate speeds, there is a rich spectrum of excited frequencies in both directions, which results in vibrations with comparatively large amplitudes that may induce fatigue failure.

Misalignment occurs when the journal’s centerline does not coincide with the bearing’s centerline. This is caused by combinations of rotational movements about a pivot point in the longitudinal cross-section and translational movements of the journal in the vertical and horizontal axis in the radial cross-section. Misalignment results from assembling or manufacturing errors, off-centric loads, shaft deflection such as elastic and thermal distortions, and externally imposed misaligned moments. A side-effect of the misalignment is the creation of a converging wedge geometry, known as the oil wedge, between the journal and the outer housing. In addition, misalignment contributes to the whirl by changing the threshold speed at which instability occurs. It is able to change the journal bearing’s load-carrying capability, increase its frictional power loss, alter the fluid film thickness, change the dynamic characteristics such as system damping and critical speeds, and modify the vibrations as well as the overall stability of the system. One of the most significant effects of misalignment is its ability to produce a substantial amount of vibration when the frequency of the rotor’s vibrations is a harmonic of the rotational speed of the journal.

Hot spots found on a journal, which ultimately develop into thermal bends, are results of the Newkirk effect. These hot spots are quite common, result-

ing from the contact of the rotor with the bearing or due to a temperature difference across the diameter of the journal. In the latter case, the temperature difference is a result of the differential shearing in the oil film. This phenomenon, in conjunction with the system running near a critical speed of the journal, can generate unstable vibrations in the bearing.

Under normal, low journal velocity conditions, the journal resides at an equilibrium position that is determined by its velocity. However, as the speed increases and approaches a threshold speed of instability, the journal’s stability becomes compromised. Speeds above the threshold speed cause a self-excited oscillation to occur, during which the whirling motion of the journal is increased by its own rotational energy. This is dangerous if the oscillations are of large magnitude. If the journal suddenly becomes unstable, this is called subcritical bifurcation. On the other hand, if the journal gradually becomes unstable, it is called supercritical bifurcation. Subcritical bifurcation is also possible under the threshold speed when the rotor is given small perturbations by an outside force. Consequently, factors such as constant and imbalance loads on the journal are considered important when preventing bifurcation.

Vibrations can also occur from the lack of an oil wedge. Oil wedges, which are responsible for the load-carrying capability of the journal bearing, can be prevented from forming if the load is too heavy, the journal speed is too slow, or there is a lack of lubrication. In all three cases, metal-to-metal contact occurs, thus causing vibrations in the bearing.

Seizure, which results in a complete halt of the journal’s movement, is a serious common problem. This mode of failure can result from the lack of an oil wedge (dry rubbing) which leads to highly localized heating, inadequate heat release from the system, and thermal expansion of the journal. In the third case, the journal may thermally expand faster than the bearing housing, causing the clearance between the journal and the bearing to disappear and metal-to-metal contact to occur. In the case of tilt pad bearings, thermal expansion of the tilt pads can produce the same phenomenon.

See also: **Balancing; Diagnostics and condition monitoring, basic concepts.**

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BEARING VIBRATIONS

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Introduction

Rotors in rotating machinery are guided by links. They are classified into three categories:

1. Fluid
2. Roller bearings
3. Magnetic field

Forces are transmitted to nonrotating parts. They are a result of associated impedance:

- To links
- To sealing systems
- To elements connected to the rotor, such as wheels.

Sealing systems and elements connected to the rotor play a part in guiding the rotor. Coupling between rotors falls into the link category. This article describes the characteristics of links.

Functions

Links must allow:

- Fixed-precision guidance to avoid rotor–stator contact
- Minimum wear and tear to reduce machine unavailability
- Removal of heat, in particular from the rotor
- Minimization of leaks
- Neutralization or removal of impurities
- Minimum loss through friction, for economy and low-level heating